

RECURSIVE LANGUAGE MODELS

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ABSTRACT

We study allowing large language models (LLMs) to process arbitrarily long prompts through the lens of inference-time scaling. We propose **Recursive Language Models (RLMs)**, a general inference strategy that treats long prompts as part of an external *environment* and allows the LLM to *programmatically* examine, decompose, and recursively call itself over snippets of the prompt. We find that RLMs successfully handle inputs up to two orders of magnitude beyond model context windows and, even for shorter prompts, dramatically outperform the quality of base LLMs and common long-context scaffolds across four diverse long-context tasks, while having comparable (or cheaper) cost per query.

1 INTRODUCTION

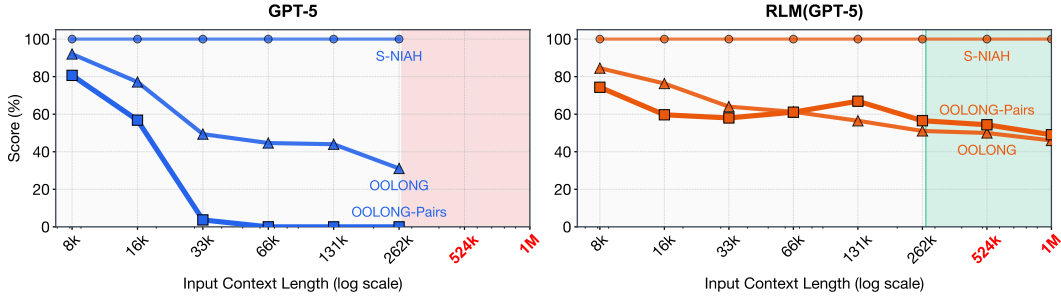


Figure 1: A comparison of GPT-5 and a corresponding RLM on three long-context tasks of increasing complexity: **S-NIAH**, **OOLONG**, and **OOLONG-Pairs**. For each task, we scale the input length from 2^{13} to 2^{18} . GPT-5 performance degrades significantly as a function of both input length and task complexity, while the RLM maintains strong performance. Inputs beyond the red region do not fit in GPT-5’s context window of 272K tokens, but the RLM handles them effectively. Additional experiments across other models, methods, and benchmarks are in §2.

Despite rapid progress in reasoning and tool use, modern language models still have limited context lengths and, even within these limits, appear to inevitably exhibit *context rot* (Hong et al., 2025), the phenomenon illustrated in the left-hand side of Figure 1 where the quality of even frontier models like GPT-5 degrades quickly as context gets longer. Though we expect context lengths to steadily rise through improvements to training, architecture, and infrastructure, we are interested in *whether it possible to dramatically scale the context size of general-purpose LLMs by orders of magnitude*. This is increasingly urgent as LLMs begin to be widely adopted for long-horizon tasks, in which they must routinely process tens if not hundreds of millions of tokens.

We study this question through the lens of scaling inference-time compute. We draw broad inspiration from *out-of-core* algorithms, in which data-processing systems with a small but fast main memory can process far larger datasets by cleverly managing how data is fetched into memory. Inference-time methods for dealing with what are in essence long-context problems are very common, though typically task-specific. One general and increasingly popular inference-time approach in this space is context condensation or compaction (Khattab et al., 2021; Smith, 2025; OpenAI, 2025; Wu et al., 2025), in which the context is repeatedly summarized once it exceeds a length threshold. Unfortunately, compaction is rarely expressive enough for tasks that require dense access